

Campus Image-to-GPS Regression for Localization and Navigation

Final Report

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1. Abstract

This report describes an end-to-end deep learning pipeline for campus-scale image geolocalization. Given a single RGB image captured on a university campus, the system predicts the absolute GPS coordinates (latitude, longitude) of the camera using a fine-tuned convolutional neural network regressor. A dataset of 1,931 campus images was collected and labeled via EXIF GPS metadata, yielding 1,517 usable annotated samples spanning both daytime and nighttime conditions. Three backbone architectures were benchmarked under identical experimental conditions: ResNet50, EfficientNet-B0, and ConvNeXt-Tiny. ResNet50 achieved the lowest error across all metrics (MDE 14.06 m, MedDE 10.58 m, Acc@5 m 17.32%, Acc@10 m 47.19%) and was selected as the final backbone. A subsequent fine-tuning stage on a domain-matched photograph split further reduced the median distance error to 4.04 m and raised 5-meter accuracy to 55.26%. The inference API exposes a `predict_gps(image)` function that returns a float32 numpy array [latitude, longitude], compliant with the course evaluation specification.

2. Introduction

Visual localization — the problem of estimating camera pose or position from an image — is a fundamental challenge in computer vision with applications in robotics, augmented reality, and autonomous navigation. In environments where satellite-based GPS signals are degraded (near complex building overhangs, indoors, or under dense vegetation), image-based localization offers a complementary mechanism that operates purely from visual content.

This project addresses a specific instance of the problem: given a single unannotated RGB photograph taken anywhere on a university campus, predict the absolute WGS-84 GPS coordinates (latitude, longitude) of the camera. The task is framed as a continuous regression problem. Unlike retrieval-based approaches that match query images to a geo-tagged database, our method performs direct coordinate regression through a pre-trained convolutional backbone combined with a lightweight fully connected regression head.

The primary challenges in this setting are:

- Fine-grained visual similarity: nearby locations share similar architectural features, making small positional differences hard to resolve from image appearance alone.
- Day/night domain shift: the same scene looks radically different under sunlight versus artificial illumination, creating an in-distribution generalization problem.
- Label noise from EXIF GPS: GPS coordinates embedded in image metadata carry real-world measurement error (typically $\pm 5\text{--}15$ m for consumer devices), creating a hard accuracy ceiling independent of model capacity.
- Limited dataset scale: with only $\sim 1,500$ labeled examples across a constrained campus area, the model must leverage transfer learning from large-scale ImageNet pre-training.

The main contributions of this work are:

- A complete EXIF-driven data collection and labeling pipeline for a self-collected campus GPS dataset (1,931 images, 1,517 labeled).
- A controlled multi-backbone benchmark (ResNet50, EfficientNet-B0, ConvNeXt-Tiny) under a shared, leak-free split manifest.
- A targeted fine-tuning protocol on a domain-matched photograph split, demonstrating a 28% reduction in median distance error using only five additional training epochs.
- A reproducible inference API — `predict_gps(image: np.ndarray) → np.ndarray[2, float32]` — that operates without hardcoded absolute paths and passes the course evaluation contract.

3. Related Work

Visual geolocalization has been approached through both retrieval and regression paradigms. Retrieval methods (e.g., NetVLAD [Arandjelovic et al., 2016]) encode query and database images into a shared descriptor space and return the GPS label of the nearest database match. These methods scale well with database size but require maintaining the full image index at inference time. Regression methods, as employed in this work, instead train a model to map images directly to coordinates, avoiding the database dependency at the cost of generalization beyond the training distribution.

PoseNet [Kendall et al., 2015] introduced direct CNN-based camera pose regression using GoogLeNet as a backbone. While originally designed for 6-DoF pose estimation, the same regression head design inspired subsequent GPS regression work. Our approach adopts a similar architecture — a frozen or partially unfrozen backbone with a two-output regression head — but focuses exclusively on 2D coordinate prediction (latitude, longitude) on a constrained campus domain.

Among the backbone architectures evaluated, ResNet50 [He et al., 2016] introduced residual skip connections that allow very deep networks to train stably. Its strong ImageNet representations have made it a reliable transfer learning foundation across diverse downstream tasks. EfficientNet [Tan and Le, 2019] proposes compound scaling of network depth, width, and input resolution under a fixed computational budget. Despite achieving state-of-the-art classification accuracy at small parameter counts, our experiments show that its benefits do not transfer to this coordinate regression setting. ConvNeXt [Liu et al., 2022] modernizes the convolutional architecture with techniques borrowed from

transformer design (larger kernels, layer normalization, GELU activation) while preserving the CNN inductive bias. Again, the architectural sophistication does not compensate for the small dataset scale in our domain.

Evaluation in visual localization is typically performed using geodetic distance rather than Euclidean coordinate difference, because the lat/lon coordinate system is non-linear at local scales. We follow the standard practice of computing Haversine distance on WGS-84 coordinates using double precision arithmetic, reporting mean distance error (MDE), median distance error (MedDE), and threshold accuracy at 5 m, 10 m, 20 m, and 50 m.

4. Method

4.1 Data Collection and Labeling

Images were collected on campus using smartphones and digital cameras across multiple sessions covering both daytime (natural light) and nighttime (artificial illumination) conditions. All images were taken in RGB color with varying resolutions depending on the capture device. GPS ground truth was extracted from the EXIF metadata embedded in each image file (fields `GPSLatitude` and `GPSLongitude` in WGS-84 decimal degrees). Images where either field was absent, zero, or outside the expected campus bounding box were discarded. After filtering, 1,517 images retained valid GPS labels (795 daytime, 722 nighttime) from the original 1,931 captured (795 daytime, 1,136 nighttime).

For reporting, the dataset is stratified only by lighting condition (day/night) rather than by collector identity, to avoid exposing personal information in the experimental analysis. The ground truth coordinate files (`photos_gt.csv`, `campus_gps_master_dataset.csv`) record image filename, latitude, and longitude, matching the format required by the course evaluation specification.

4.2 Input and Output Representation

Input: A single RGB image with shape (H, W, 3), uint8 values in [0, 255]. The resolution varies by capture device; all images are resized to 224×224 pixels (240×240 for EfficientNet variants) during preprocessing.

Output: A numpy array of shape (2,) and dtype float32, where `output[0]` is the predicted WGS-84 latitude and `output[1]` is the predicted WGS-84 longitude. This signature matches the course evaluation contract: `predict_gps(image: np.ndarray) → np.ndarray`.

4.3 Model Architecture

All models follow a two-component design: a pre-trained convolutional backbone for feature extraction, and a task-specific regression head that maps backbone features to GPS coordinates.

The backbone is initialized from ImageNet pre-trained weights (torchvision). The final classification layer is removed and replaced with a regression head consisting of a global average pooling layer (if not already present), followed by a `BatchNorm1d` layer, a `Linear(feature_dim → 256)` layer with ReLU activation and 0.3 dropout, and a final `Linear(256 → 2)` output layer. The output is in scaled coordinate space (see Section 4.4); inverse transformation recovers physical lat/lon coordinates.

Three backbone families were evaluated:

- ResNet50: 25.5 M parameters, feature dimension 2048. Strong residual skip connections provide stable gradient flow across 50 layers. Selected as final backbone.
- EfficientNet-B0: 5.3 M parameters, feature dimension 1280. Compound-scaled architecture balances depth, width, and resolution. Training was halted by early stopping at epoch 72.
- ConvNeXt-Tiny: 28.6 M parameters, feature dimension 768. Modernized convolutional design inspired by Vision Transformers. Training halted at epoch 59.

4.4 Training Procedure

All training followed the same protocol to ensure fair comparison. The dataset was split using a fixed random seed (seed=42) into train (70%), validation (15%), and test (15%) partitions. The exact image-level split indices were serialized to a shared JSON manifest (finetune_shared_split_photos_seed42.json, finetune_shared_split_seed42.json), ensuring that all backbone runs are compared on identical val and test sets.

Target scaling. Raw GPS coordinates were normalized using `MinMaxScaler(feature_range=(-1, 1))`, fit exclusively on the training split and then applied to val and test. The fitted scaler is persisted as a .pkl file alongside each model checkpoint. Fitting after splitting prevents the validation and test coordinate ranges from leaking into the normalization parameters, which would artificially tighten the loss scale on those partitions.

Loss function. Huber loss with $\delta=0.1$. Huber loss is quadratic for errors below δ and linear above, making it more robust to outlier predictions than mean squared error. The small δ value (0.1 in scaled coordinate space, corresponding to a few tens of meters in physical space) ensures that catastrophic night-scene outliers do not dominate the gradient signal during training.

Optimizer. Adam with differential learning rates: the backbone parameters are updated at $0.1 \times \text{head_lr}$ (1×10^{-4}), while the regression head is updated at $\text{head_lr} = 1 \times 10^{-3}$. This two-rate scheme allows the head to adapt quickly while the pre-trained backbone features evolve more conservatively, preventing destructive forgetting of ImageNet representations.

Scheduler. ReduceLROnPlateau with factor=0.5 and patience=10, monitoring validation loss. Early stopping with patience=12 terminates training when validation loss fails to improve for 12 consecutive epochs. Gradient clipping ($\text{max_norm}=1.0$) prevents exploding gradients.

Augmentation. Training images are augmented with `RandomHorizontalFlip`, `ColorJitter(brightness=0.4, contrast=0.4, saturation=0.3, hue=0.15)`, `RandomGrayscale(p=0.2)`, `GaussianBlur`, `RandomAdjustSharpness`, `RandomRotation(degrees=10)`, and ImageNet normalization (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Validation and test images receive only resize and normalization.

Test-time augmentation (TTA). At inference, three variants of each image are passed through the model: (1) the original image, (2) a horizontal flip, and (3) a brightness-boosted version (brightness factor 1.3). The three predicted coordinate vectors are averaged in scaled space before inverse transformation. TTA reduces the variance of individual predictions and is particularly effective for images near decision boundaries in the campus map.

4.5 Evaluation Metrics

All errors are computed as Haversine distances on WGS-84 coordinates using the Earth mean radius $R = 6,378,137$ m and float64 arithmetic. The following metrics are reported:

- **Mean Distance Error (MDE):** arithmetic mean of per-image Haversine errors in meters. Sensitive to large outliers; reflects worst-case system behavior.

- Median Distance Error (MedDE): median of per-image Haversine errors. Robust to outliers; represents the typical navigation experience for a standard user.
- Accuracy at threshold (Acc@K m): fraction of test images with predicted error $\leq K$ meters, for $K \in \{5, 10, 20, 50\}$.

5. Experiments

5.1 Backbone Benchmark

The first experiment compares the three candidate backbones on an identical experimental setup. All models were trained for up to 100 epochs on the full labeled dataset (campus_gps_master_dataset.csv, $N \approx 1,540$) using the shared split manifest (seed=42, val=15%, test=15%, $N_{\text{test}}=231$). Early stopping may terminate training before 100 epochs if validation loss stagnates.

Table 1 summarizes the results on the held-out test split.

Model	Epochs	MDE (m)	MedDE (m)	Acc@5m	Acc@10m	Acc@20m
ResNet50 ★	100	14.06	10.58	17.32%	47.19%	79.22%
EfficientNet-B0	72 ‡	17.59	14.10	12.99%	35.50%	73.16%
ConvNeXt-Tiny	59 ‡	15.87	11.49	17.32%	42.86%	76.19%
★ Selected backbone ‡ Stopped early by patience=12						

Table 1. Backbone benchmark results on the held-out test split ($N=231$). All models trained under identical conditions using the shared split manifest.

ResNet50 achieves the lowest MDE (14.06 m), MedDE (10.58 m), and highest Acc@10m (47.19%). It is the only model to complete the full 100 epochs without early stopping, demonstrating stable optimization dynamics. EfficientNet-B0 triggered early stopping at epoch 72; despite its compact architecture, it underperforms on all metrics, suggesting that compound scaling does not provide an advantage for this regression task with limited data. ConvNeXt-Tiny halted at epoch 59; its test MDE (15.87 m) is 1.81 m higher than ResNet50 despite a larger parameter count (28.6 M vs 25.5 M).

5.2 Final Fine-Tuning

To isolate the contribution of the fine-tuning stage, the best backbone (ResNet50) was further trained on the domain-matched photographs split (photos_gt.csv, $N \approx 1,267$), using a tighter shared split manifest (seed=42, val=15%, test=15%, $N_{\text{test}}=190$). The fine-tuning run (resnet50_20260315_171307) initialized from the best checkpoint of the baseline run (resnet50_20260305_2009) and trained for up to 20 epochs with early stopping patience=12. The best validation checkpoint was reached after epoch 5.

Table 2 shows the before/after comparison on the photographs test split.

Configuration	MDE (m)	MedDE (m)	Acc@5m	Acc@10m	Acc@20m
Baseline (no FT)	9.17	5.65	45.26%	74.21%	88.95%
Fine-tuned (5 ep)	7.64	4.04	55.26%	80.53%	91.05%
Δ (gain)	-1.53 m	-1.60 m	+10.00 pp	+6.32 pp	+2.10 pp

Table 2. Before/after fine-tuning comparison on the photographs test split (N=190). Δ (gain) is fine-tuned minus baseline; negative values indicate improvement.

Fine-tuning for 5 epochs on the matched split reduces MDE by 1.53 m (−16.7%) and MedDE by 1.60 m (−28.3%). The 5-meter accuracy improves by 10 percentage points, meaning one in ten additional test images is now localized within a pedestrian-scale radius. The rapid convergence (best checkpoint at epoch 5) confirms that the baseline model was already well-calibrated and the fine-tuning stage primarily corrects the coordinate distribution shift between the two dataset splits.

5.3 Validation and Training Dynamics

The baseline run (resnet50_20260305_2009) trained stably to epoch 100 with the best validation checkpoint reached at epoch 88 (val_loss=0.007958, val_MDE=16.14 m, val_MedDE=10.09 m, val_Acc@5m=22.51%). The validation loss remained within a narrow band after epoch 70, indicating that the model had converged. The fine-tune run (resnet50_20260315_171307) improved validation MDE from 9.17 m at epoch 0 to 6.11 m at epoch 5 (best), with best val_loss=0.001856.

The minimum test error per run confirms the absence of overfitting on the test set: the minimum single-image error for the final model is 0.42 m, demonstrating that the network can achieve sub-meter accuracy on well-represented, well-lit campus views.

6. Ablation Study

The method consists of three non-trivial components beyond a vanilla fine-tuning baseline: (A) the choice of backbone architecture, (B) the inclusion of a targeted fine-tuning stage, and (C) the isolated scaling protocol. Each is evaluated below.

6.1 Ablation A: Backbone Architecture

To assess the necessity of using ResNet50 as the backbone, we compare the final test metrics across all three evaluated architectures on the same backbone benchmark split (N=231). All other hyperparameters and the split manifest are held fixed.

As shown in Table 1, replacing ResNet50 with EfficientNet-B0 degrades MDE by +3.53 m (a relative increase of +25.1%) and drops Acc@5m by 4.33 percentage points. Replacing with ConvNeXt-Tiny increases MDE by +1.81 m and reduces Acc@10m by 4.33 pp. In all metrics, ResNet50 is the dominant choice. The result is attributable to two factors: (1) ResNet50's 2048-dimensional feature space provides a richer representation pool for the regression head compared to ConvNeXt-Tiny's 768-dimensional output, and (2) EfficientNet-B0's early stopping suggests optimization instability on this small dataset, possibly due to its compound-scaled normalization layers interacting poorly with the narrow coordinate regression loss landscape.

Conclusion: backbone choice is a necessary component. Removing ResNet50 and using any evaluated alternative increases both mean and median error.

6.2 Ablation B: Fine-Tuning Stage

To quantify the contribution of the fine-tuning stage, we compare the ResNet50 model evaluated directly on the photographs test split (N=190) — without any fine-tuning — against the fine-tuned version. The baseline numbers correspond to the resnet50_20260315_170940 evaluation run, which applies the baseline checkpoint to the photos split without any additional training.

Without fine-tuning: MDE 9.17 m, MedDE 5.65 m, Acc@5m 45.26%. With fine-tuning (5 epochs): MDE 7.64 m, MedDE 4.04 m, Acc@5m 55.26%. The fine-tuning stage provides a consistent and statistically meaningful improvement across every metric. The MedDE reduction of 1.60 m (28.3%) from just five gradient descent epochs is particularly striking and reflects strong prior initialization from the 100-epoch baseline training.

Conclusion: the fine-tuning stage is a necessary and highly effective component. Removing it degrades the median user experience by 1.60 m and drops 5-meter accuracy by 10 percentage points.

6.3 Ablation C: Isolated Scaling (No Data Leakage)

A subtle but critical design choice is fitting the MinMaxScaler exclusively on the training split rather than on the entire dataset. Fitting on the full dataset would allow the coordinate range of the validation and test sets to inform the normalization, constituting a form of label leakage. While such leakage does not directly cause the model to see test labels during gradient updates, it changes the effective target range

seen during training and invalidates the claim of a fair cross-backbone comparison (since different dataset splits could have different coordinate ranges in each fold).

We did not run a controlled ablation by deliberately leaking the scaler, as this would require re-training all three backbones with a contaminated scaler. Instead, we note that the academic literature consistently reports inflated performance when preprocessing statistics are computed on the full dataset [Kapoor and Narayanan, 2022]. Our protocol guards against this by using a strict split-then-fit ordering and a shared manifest.

7. What Did Not Work

Several approaches were explored or considered during development that did not produce useful results or were abandoned based on preliminary evidence.

7.1 Nighttime Image Recovery

A large portion of the captured images (1,136 nighttime) lacked EXIF GPS metadata, reducing the usable labeled night set to 722 images. Several approaches to recover or synthetically generate GPS labels for the unlabeled night images were considered, including matching night images to day images of the same location by manual annotation and SIFT-based correspondence. These approaches were abandoned due to the labeling effort required and the risk of introducing systematic errors through imperfect matching.

7.2 Early Cross-Run Benchmark Comparability

Initial benchmark runs were conducted before the shared split manifest was established. Each backbone was trained on independently drawn random splits, making direct metric comparison unreliable (different test sets could have different difficulty profiles). These early results were discarded and all three backbones were re-evaluated from scratch under the shared manifest protocol.

7.3 Larger Regression Head Depth

Preliminary experiments with a three-layer regression head (backbone $\rightarrow 512 \rightarrow 256 \rightarrow 2$) showed no improvement over the two-layer design (backbone $\rightarrow 256 \rightarrow 2$) and introduced additional optimization instability. The deeper head was abandoned in favor of the simpler architecture with stronger regularization (BatchNorm + Dropout).

7.4 Night-Only Training Subset

A brief experiment trained a model exclusively on nighttime images to assess whether a specialized night model would outperform the mixed-condition baseline on nighttime test images. The night-only model converged to higher error than the mixed model, likely due to the smaller training set size (722 images vs $\sim 1,065$ in the mixed training split) and the increased noise in nighttime feature representations.

8. Discussion and Limitations

8.1 Interpretation of Results

The final model achieves a median distance error of 4.04 m. For perspective, consumer GPS receivers typically report accuracy of $\pm 3\text{--}10$ m under open-sky conditions. The median performance of the system is therefore within the range of hardware GPS, which is encouraging for a vision-only regressor trained on a dataset of 1,517 images. However, the mean distance error (7.64 m) is inflated by a long tail of high-error predictions, predominantly from nighttime images where sensor noise destroys the edge and texture cues that the backbone relies on.

The gap between MDE (7.64 m) and MedDE (4.04 m) implies a skewed error distribution: the majority of predictions are accurate, but a small fraction of catastrophic failures (errors > 30 m) disproportionately

inflate the mean. Addressing this tail — through better night data, uncertainty estimation, or outlier rejection — would bring MDE much closer to MedDE without necessarily improving the median further.

8.2 Limitations

The labeled dataset is small by deep learning standards. With only ~1,500 samples, the model is heavily dependent on ImageNet pre-training and cannot learn campus-specific features that are not already partially encoded in the backbone weights. Collecting more images — particularly under diverse lighting conditions and from underrepresented campus regions — would likely yield substantial gains.

The system does not produce uncertainty estimates. A model that could flag predictions as high-uncertainty (e.g., from featureless walls or dark corridors) would be safer to deploy in a real navigation context, since it could fall back to alternative localization mechanisms when confidence is low.

The `predict_gps` function currently requires the model checkpoint path to be configured relative to the repository root. Absolute path configuration would improve portability across different deployment environments.

8.3 Future Directions

Based on the failure analysis, the most impactful immediate improvements would be: (1) oversampling nighttime images in the fine-tuning stage or applying domain-adversarial training to close the day/night domain gap; (2) adding a confidence head to the regression model that predicts expected error alongside the coordinate estimate, enabling uncertainty-aware inference; (3) exploring scene-type conditioning (structural vs. open-air) to provide the model with prior knowledge about the expected error magnitude.

9. References

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